

VASANTHA NAGESWAR REDDY MADAKALA | Sr. Data Analyst

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SUMMARY

Results-driven Senior Data Analyst / Data Engineer with **5+** years of experience designing enterprise analytics solutions, automating data pipelines, and delivering high-value insights across **cloud, healthcare, SaaS, and digital-transformation** environments. Expert in building scalable **ETL/ELT** workflows using **Apache Airflow, ADF, AWS Glue, SQL, and PySpark**, with a strong track record of reducing manual workload by **30–40%** and improving data availability and reliability across distributed systems. Adept at developing high-impact dashboards in **Power BI, Tableau, and Azure Data Explorer** to drive executive-level decision-making, operational efficiency, and KPI visibility. Skilled in predictive modeling, statistical analysis, and experiment design using **Python, R, and scikit-learn**-enabling measurable improvements in forecasting accuracy, retention strategy, and product performance validation. Experienced in data quality engineering, post-migration validation, schema modeling, and real-time processing using **Spark, Kafka, Snowflake, BigQuery, and SQL Server**. Proven collaborator across engineering, product, IT, and business teams, delivering insights that have supported multimillion-dollar funding proposals, optimized supply chain and operational workflows, and strengthened enterprise cloud and cybersecurity monitoring

TECHNICAL SKILLS

Data Engineering & ETL	Azure Data Factory (ADF), Apache Airflow, AWS Glue, ETL/ELT Pipelines, Data Quality Frameworks, Data Validation & Reconciliation, Data Cleaning & Transformation, Cross-System Workflows, Cloud Data Migration
Databases & Warehousing	Azure SQL, SQL Server, BigQuery, Snowflake, PL/SQL, Star & Snowflake Schema Modeling, Cloud-Native Architecture, Distributed Computing
Analytics & BI Tools	Power BI (DAX, Power Query, Data Modeling), Tableau (Geospatial Analytics), Salesforce CRM, Google Analytics, KPI Frameworks, Stakeholder Reporting
Programming & Data Science	Python (Pandas, NumPy, Scikit-learn, PySpark, Matplotlib, Seaborn), R (Statistical Modeling, A/B Testing), Spark, Kafka, Advanced Excel (VBA), Machine Learning (Forecasting, Churn Models), Statistical Analysis
Cloud & Platforms	Azure Cloud, Multicloud Infrastructure, Microsoft Dataverse, Power Apps, Power Automate, API Integrations, Real-Time Data Processing
Tools & Project Methodologies	Git, JIRA, Rally, Agile, Scrum, Waterfall

PROFESSIONAL EXPERIENCE

IBM Armonk, NY	March 2025 – Present
Sr. Data Analyst	
- Partnered with engineering, analytics, and product stakeholders to capture complex business requirements and translate them into scalable dashboards in Power BI and Azure Data Explorer, incorporating enterprise KPIs and actionable insights for operational and strategic decision-making. - Performed large-scale post-migration data reconciliation across Azure cloud systems, resolving 200+ discrepancies between legacy SQL Server sources and cloud datasets through advanced SQL joins, window functions, and PySpark-based validation routines. - Designed and deployed automated data quality frameworks in Azure Data Factory, integrating rule-based validation, logging, and exception handling to reduce manual reconciliation workloads by 40% and ensure enterprise-level data reliability. - Built high-impact dashboards using Power BI for trend analysis and Tableau for geospatial visualization, delivering program-level KPI insights that improved stakeholder decision-making turnaround by 30%. - Automated end-to-end data extraction, cleansing, and validation using Python workflows orchestrated via Apache Airflow, reducing manual processing by 40 hours monthly and ensuring compliance with federal energy-efficiency reporting standards. - Developed enterprise-grade Power BI and Tableau dashboards to monitor financial performance, operational KPIs, and grant utilization, equipping leadership teams with real-time visibility to accelerate business planning cycles. - Engineered automated ETL pipelines using Apache Airflow and AWS Glue, optimizing dependency handling and refresh cycles to improve daily data availability by 18% for analytics and reporting teams. - Built predictive machine learning models using scikit-learn for churn prediction and sales forecasting, improving retention insights and contributing to a 9% increase in customer retention within a single quarter. - Designed and executed controlled A/B experiments using R for statistical testing (t-tests, ANOVA, regression diagnostics), enabling product teams to validate new feature performance with statistically significant results. - Performed deep-dive statistical and predictive analysis using Python (Pandas, NumPy, and Scikit-learn), uncovering trends that supported 10–20% operational improvement across targeted processes. - Built and maintained scalable data models, including star/snowflake schemas in SQL Server/BigQuery/Snowflake, enhancing reporting performance and reducing query time by 60%. - Conducted advanced cohort analyses and energy-consumption forecasting using Python (scikit-learn) and Excel VBA automation, generating quantifiable insights that supported \$500K+ in successful funding proposals for clean-energy and digital-equity initiatives.	

Data Analyst

- Designed and automated end-to-end ETL pipelines using **Apache Airflow** for orchestration combined with SQL-based transformations, leveraging advanced scheduling and dependency management to save **30+ hours** per month in claims processing and ensure timely delivery of analytics-ready datasets.
- Developed interactive analytics solutions by combining **Power BI** for enterprise-level dashboards with **Python (Matplotlib/Seaborn)** for customized visual insights, accelerating decision-making efficiency by **20%** for clinical, actuarial, and IT business units.
- Defined enterprise data requirements and integrated **Microsoft Dataverse** with **Power Apps, Power Automate, and external APIs** to create seamless cross-system workflows.
- Developed high-performance solutions for real-time distributed data processing and computational pipelines leveraging **Spark, PySpark**, and cloud-native architecture.
- Authored optimized **PL/SQL** and **SQL queries** to extract, aggregate, and transform large operational datasets, supporting precise reporting for finance, compliance, and enterprise performance teams.
- Cleaned, transformed, and standardized high-volume datasets using **Python** and **R**, reducing data inconsistencies and lowering quarterly reporting discrepancies by **18%** across multiple business functions.
- Built **Tableau dashboards** to visualize multi-year regional sales trends, enabling leadership to identify performance gaps and drive more strategic resource allocation at the national level.
- Collaborated with data engineering teams to integrate **Kafka-based streaming pipelines**, enabling real-time ingestion and continuous monitoring of operational metrics across business platforms.
- Developed and tuned machine learning models for demand forecasting using **Python and scikit-learn**, improving forecast accuracy by **9%** and directly enhancing supply chain responsiveness.
- Supported **Agile** and **Waterfall** project delivery using **Rally**, documenting technical and functional requirements, participating in sprint ceremonies, and ensuring alignment between data initiatives and organizational objectives.

CitiusTech | India

January 2019 - December 2021

Data Analyst

- Engineered and supported enterprise dashboards across Power BI, Tableau, and SAP BusinessObjects, applying advanced DAX, LOD expressions, and semantic data modeling to provide real-time visibility into IT infrastructure KPIs for 100+ global users, reducing reporting latency by **15%**.
- Designed and executed ETL workflows processing **5M+ records per month** using SQL Server and Python (Pandas), producing scalable, high-quality datasets that enhanced multi-cloud operational analytics and strengthened cybersecurity monitoring across distributed environments.
- Automated end-to-end data workflows by integrating Python transformation scripts with Apache Airflow DAG orchestration, improving pipeline reliability, enabling automated alerting and dependency handling, and reducing manual intervention by **20%** while ensuring daily data refresh SLAs.
- Elevated supply chain optimization efforts by analyzing procurement, logistics, and inventory datasets using SQL and Python; built Power BI dashboards that increased supplier visibility and inventory accuracy, improving order-fulfilment rates by **18%** and reducing surplus inventory by **10%**.
- Collaborated with IT, cybersecurity, and product engineering teams within Agile/Scrum frameworks using JIRA to drive sprint planning, backlog refinement, and cross-team issue management-contributing to a **12% improvement in server uptime** across hybrid and multi-cloud platforms.
- Performed statistical analysis and implemented machine learning models (scikit-learn) to enable proactive anomaly detection for network traffic and infrastructure events, improving detection accuracy across **300+ critical nodes** and reducing mean-time-to-identify (MTTI) service issues.
- Developed marketing analytics dashboards by integrating data from Salesforce CRM, Google Analytics, and multi-channel campaign platforms, enabling full-funnel performance visibility and contributing to a **20% increase in marketing ROI**.
- Delivered executive-level reporting and strategic insights through advanced Excel models (including VBA automation) and PowerPoint analytics summaries, translating complex financial risk models into recommendations that influenced **\$2M+ investment decisions** in next-generation infrastructure technologies.

EDUCATION

- **M.S. in Data Science | Sep 2023 – May 2025** Saint Peter's University
- **Bachelor's in computer science and engineering | Jul 2016 - May 2020** Siddharth Institute of Engineering and Technology

CERTIFICATION

- **Microsoft Certified: Azure Data Fundamentals | Feb 2025**
- **IBM Artificial Intelligence Fundamentals | Dec 2024**